
MA 609: Numerics of Partial Differential Equations

Course Code	MA 609
Course Name	Numerics of Partial Differential Equations
L-T-P-C	3-0-0-3
Prerequisites	MA 607 (Numerical Analysis); Knowledge in Differential Equations
Intended for	M.S./Ph.D, B.Tech. 3rd and 4th year students
Distribution	Elective
Approval	4th Senate (May 2013)

Course Contents

- **Introduction to Finite difference schemes** [5 lectures]
Finite difference schemes for partial differential equations, explicit schemes, implicit schemes, single step schemes, multi-step schemes.
- **Finite difference schemes for boundary value problems** [10 lectures]
FTCS, backward Euler and Crank-Nicolson schemes, ADI methods, Lax-Wendroff method, upwind scheme.
- **Consistency, stability and convergence Analysis** [10 lectures]
Stability analysis by von Neumann method, CFL condition, Lax's equivalence theorem.
- **Introduction to Finite element method** [7 lectures]
Finite element method for partial differential equations, variational methods, method of weighted residuals.
- **Finite element discretization and error analysis** [10 lectures]
Finite element discretizations for one-dimensional and two-dimensional elliptic equations, a priori and a posteriori error estimates.

Text Books

1. G. D. Smith, **Numerical Solutions to Partial Differential Equations**, 3rd Edition, Oxford University Press, 1986.
2. C. Johnson, **Numerical Solution of Partial Differential Equations by the Finite Element Method**, Dover Publications, 2009.

References

1. J. C. Strikwerda, **Finite Difference Schemes and Partial Differential Equations**, SIAM, 2004.
2. E. Suli, **Finite Element Methods for Partial Differential Equations**, University of Oxford, 2000.
3. P. Niyogi, S. K. Chakrabarty, M. K. Laha, **Introduction to Computational Fluid Dynamics**, Pearson Publications, 2011.
4. J. N. Reddy, **An Introduction to Finite Element Method**, 3rd Edition, McGraw Hill, 2005.